Will Kwan

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B.Sc. graduate in Earth and Environmental Sciences with a Computer Science minor, seeking an entry-level GIS or environmental analyst role in natural resource management. Coursework spanning ecology, earth systems, and forest/watershed management, paired with applied experience in spatial and raster analysis, remote-sensing classification, and Python geospatial pipelines.

Skills

- **GIS and spatial data:** ArcGIS Pro, ArcGIS Online, geoprocessing (spatial joins, overlay, buffer analysis), feature classes and geodatabases, map projections and coordinate systems
- **Remote sensing and raster analysis:** Satellite imagery, supervised classification, raster analysis, accuracy assessment
- **Programming and data:** Python (pandas, geopandas, numpy), R, SQL, Git, Microsoft Excel; spatial and tabular data QA/QC, validation, statistics, documentation
- **Field and additional:** Water-quality and environmental field sampling; technical writing, valid Class 5 BC driver's license

Projects

Applied GIS Projects (GISC 480) — ArcGIS Pro 2025

- Georeferenced geological survey data with a 0.46 m RMS error and intersected classified surficial material units with road-network data across a 1,151 ha watershed, identifying 150 m of high-risk and 15.07 km of medium-risk roadway segments.
- Performed a supervised classification of imagery for downtown Kelowna using training polygons and building-footprint data; created 120 validation points and an error matrix, achieving 88.33% overall accuracy.

Walkability Index for the City of Kelowna — ArcGIS Pro, Python 2025

- Built a 0–100 walkability index from five indicators using open data; geocoded, cleaned, and validated 1,569 amenity addresses with Python and the Google Geocoding API.

Greyback Lake Vegetation Change Analysis — Python, Streamlit 2026

- Built a pipeline that retrieves Landsat imagery through a STAC API, applies cloud masking, creates multi-temporal composites, and computes NDVI and NBR; used Streamlit to visualize vegetation change over time between user-selected time periods.

Education

University of British Columbia, Okanagan — Kelowna, BC May 2026

Bachelor of Science, Major in Earth and Environmental Sciences, Minor in Computer Science — Cumulative average: 91.4%

Prizes and Awards:

- Kelowna Geology Committee Prize (2023–24)
- Engineers and Geoscientists BC Award in Geoscience – Okanagan (2025–26)

Relevant coursework: Fundamentals of GIS I and II, Practical Applications in GIS, Modelling and Simulation, Introductory Forest Science and Management, Management of Forested Watersheds

Research Experience

Aquatic Biogeochemistry Lab, University of Lethbridge — Lethbridge, AB

Research Assistant May 2022 – Aug 2022

- Independently conducted field sampling of water-quality and environmental parameters in wetlands, lakes, and rivers across southern Alberta, recording GPS point locations associated with samples.
- Designed, constructed, and deployed ebullitive-gas sampling traps; performed spectrophotometry and fluorometry measurements.
- Processed and organized datasets in Microsoft Excel and R using the StarDOM package.

Additional Experience

Big Fat Lion Bakeshop, *Baker* — Kelowna, BC May 2023 – Sep 2023; May 2025 – Sep 2025

Arc'teryx, *Product Guide* — Kelowna, BC May 2024 – Sep 2024